Zen, Gym, or Prayer?
Unpacking What Really Reduces Student Stress

Summary:

This paper examines how lifestyle habits influence stress levels among university students. Using data from ICADE students, a regression model was developed with variables such as sports participation, mindfulness, church attendance, pet ownership. The results show that sports participation significantly reduces stress, supporting the related hypothesis. Surprisingly, owning more digital devices is also associated with lower stress levels. Other variables, including meditation and religious activity, showed no significant effect. These findings highlight the role of physical activity in stress management and suggest unexpected effects of technology use. Keywords: stress, students, lifestyle, sports, mindfulness, digital devices.

1. Introduction

University life is often described as an opportunity of personal growth, liberation and exiting moments. Nevertheless, behind the surface there is a complicated picture. For many, this period is marked by a simple word, stress, that results as the search for stability, identity and academic pressure. Stress, in its various forms, has become a near-constant companion. And while it's widely accepted as a normal part of student life, prolonged or unmanaged stress can seriously impact both academic performance and well-being, in both the short term and long term.

In this report we will explore the subtle links between lifestyle habits, personal context, and self-reported stress indicators among students. More specifically we want to see the effects of different activities such as sports, meditation, or religious activity and how they affect stress. At the same time, we will examine how certain structural or social factors, such as number of people in the household, pet ownership, or even the number of digital devices a student owns, may help support or fight stress levels.

2. Hypothesis

Our hypotheses are:

- **Hypothesis 1 (main):** Those who practice more mindfulness have a lower percentage of stress, because they are more present and reflect more.
- **Hypothesis 2 (secondary):** Those who spend more hours doing sports have a lower percentage of stress, because they lease more cortisol and produce more oxytocin.
- **Hypothesis 3 (secondary):** Those who attend more church sessions have a lower percentage of stress, because they spend more time with the Lord.

3. Data, Sources, Model, and Variables

3.1 Data and Sources

The date used for our research comes from “Database Models 2024-25” which consists of observations from ICADE students collected through an online survey made during the academic year. The database was given to us by the professor.

The database has both quantitative and qualitative variables which vary across many different lifestyle habits, stress level and social environment. We selected variables that relate especially with stress-level for our research, and we also tried to select explanatory variables for these stress-levels (for example meditation, physical activities, prayer…).

The dataset is cross sectional, and we didn’t detect any significant losses due to missing data.

3.2 Model

The econometric model estimated to test the hypothesis is:

$$\begin{aligned} Stress\ Level\ = \ & B_1 + B_2 \times Church\ Attendance\ + \ B_3 \times Exercise\ + \ B_4 \times Mindfulnes \\ & + B_5 \times Family\ Members\ + B_6 \times Dogs\ + B_7 \times Cats + B_8 \times Romantic\ Partner \\ & + \ B_9 \times Irritability\ + \ B_{10} \times Deadlines\ + B_{11} \times Time\ Underestimation \\ & + B_{12} \times Divices\ owned \end{aligned}$$

Where Stress Level is the explanatory variable and mindfulness, exercise, church attendance, ect. are the explanatory variables as detailed in the following sections. This model seeks to test our 3 hypotheses and test how personal habits and contexts impact stress.

3.3 Variables used

This table describes the variables used in the model: it indicates their type, measurement form and impact.

Variable	Type	Measurement	Transformation	Expected sign
Stress Level	Discrete quantitative	Scale 0-100%	Dependent variable	-
Church Attendance	Discrete quantitative	No. of times attended (yearly)	-	-
Exercise	Continuous quantitative	Weekly hours	-	-
Mindfulness	Continuous quantitative	Weekly minutes	-	-
Family Members	Discrete quantitative	No. of people in the household	-	+
Dogs	Discrete quantitative	No. of dogs	-	-
Cats	Discrete quantitative	No. of cats	-	-
Romantic Partner	Binary qualitative	0=No ; 1=Yes	-	-
Irritability	Ordinal qualitative	Likert scale (0-3)	-	+
Deadlines	Ordinal qualitative	Likert scale (0-3)	-	-
Time Underestimation	Ordinal qualitative	Likert scale (0-3)	-	+
Devices Owned	Discrete quantitative	No. of personal digital devices	-	+

3.4 Descriptive Summary & Analysis

The following are the **main descriptive statistics** of the variables included in the model:

Variable	Mean	Median	Std	Min	Max	Count
MASS	31.025	15.000	46.191	0.0000	365.000	366
SPORTHOURS	4.6484	4.0000	3.4239	0.0000	25.000	366
RELAX	39.639	15.000	80.462	0.0000	600.000	366
HOUSEHOLDMEMBERS	3.9098	4.0000	1.6733	0.0000	10.000	366
DOGS	0.41257	0.0000	0.72660	0.0000	7.0000	366
CATS	0.11202	0.0000	0.59313	0.0000	8.0000	366
COUPLE	0.31421	0.0000	0.46483	0.0000	1.0000	366
ANNOYED	1.1202	1.0000	0.98858	0.0000	3.0000	366
STRESS	37.917	38.889	22.338	0.0000	94.444	366
DEADLINES	2.6530	3.0000	1.1455	0.0000	4.0000	366
TIMEUNDERSTATEMENT	2.0902	2.0000	1.0778	0.0000	4.0000	366

We are missing the percentiles (25%, 50% and 75%) as mentioned in the mock model. On gretl we only found the 5% and 95%.

Matrix of correlations between variables:

	STRESS	MASS	SPORTHOURS	RELAX	HOUSEHOLDMEMBERS	DOGS	CATS	COUPLE	ANNOYED	DEADLINES	TIMEUNDERSTATEMENT	DEVICES
STRESS	1.0	0.1	–0.2	–0.0	–0.0	–0.0	0.0	0.1	0.7	0.1	0.1	0.1
MASS	0.1	1.0	–0.2	0.2	0.2	–0.0	–0.1	0.0	0.1	0.1	–0.1	–0.0
SPORTHOURS	–0.2	–0.2	1.0	–0.0	–0.1	0.0	–0.0	0.0	–0.1	–0.1	0.0	–0.1
RELAX	–0.0	0.2	–0.0	1.0	0.1	–0.0	–0.0	–0.0	–0.0	0.0	–0.1	0.2

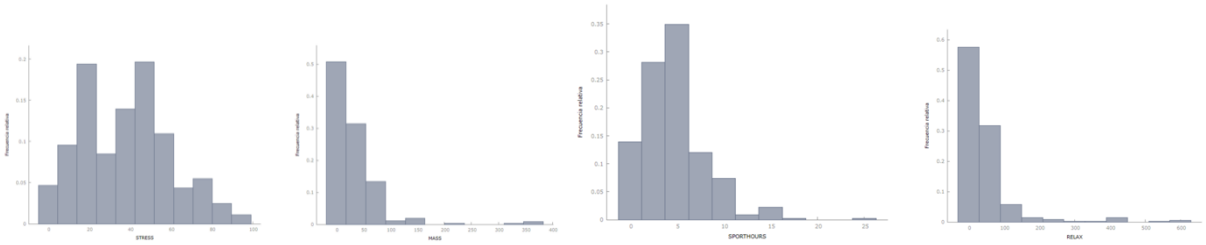
HOUSEHOLDMEMBERS	–0.0	0.2	–0.1	0.1	1.0	0.2	0.0	0.1	0.0	–0.1	–0.1	–0.0
DOGS	–0.0	–0.0	0.0	–0.0	0.2	1.0	0.1	–0.1	–0.0	–0.1	–0.1	–0.1
CATS	0.0	–0.1	–0.0	–0.0	0.0	0.1	1.0	–0.1	0.0	0.0	0.1	–0.1
COUPLE	0.1	0.0	0.0	–0.0	0.1	–0.1	–0.1	1.0	0.1	0.1	–0.1	–0.0
ANNOYED	0.7	0.1	–0.1	–0.0	0.0	–0.0	0.0	0.1	1.0	0.1	0.1	0.1
DEADLINES	0.1	0.1	–0.1	0.0	–0.1	–0.1	0.0	0.1	0.1	1.0	–0.1	–0.0
TIMEUNDERSTATEMENT	0.1	–0.1	0.0	–0.1	–0.1	–0.0	0.1	–0.1	0.1	–0.1	1.0	–0.1
DEVICES	0.1	–0.0	–0.1	0.2	–0.0	–0.1	–0.1	–0.0	0.1	–0.0	–0.1	1.0

3.4.1 Descriptive summary of the relevant variables and graphs

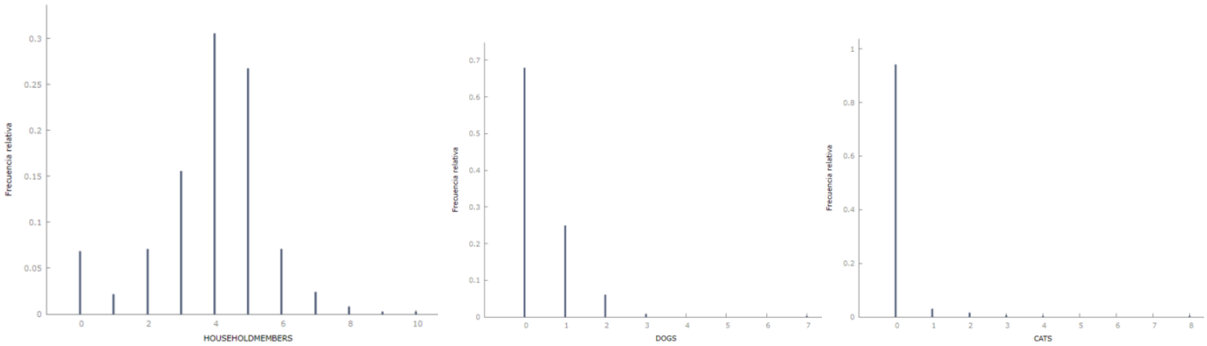
Presented below are the histograms of the main chosen explanatory variables, as well as the scatter plots for each x and y variables in addition to the correlation matrix between the explanatory variables.

Distribution of the dependent variable (STRESS) and some explanatory variables:

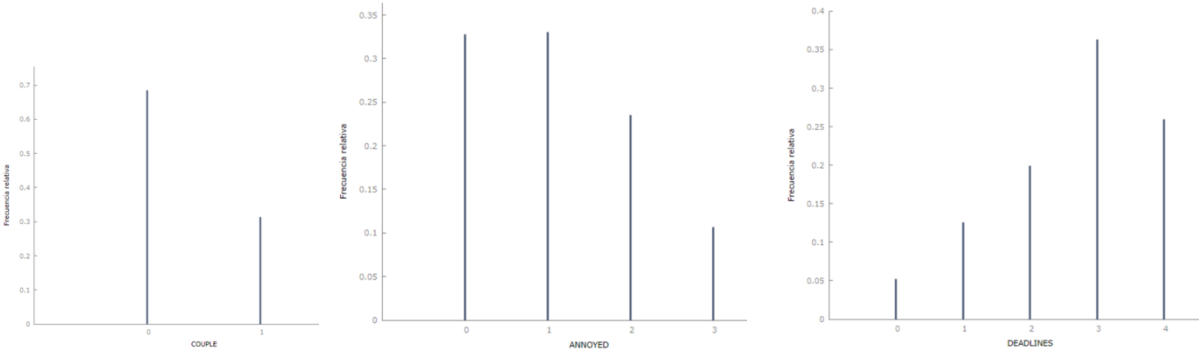
STRESS DISTRIBUTION: MASS DISTRIBUTION: SPORTHOURS DISTRIBUTION: RELAX DISTRIBUTION:



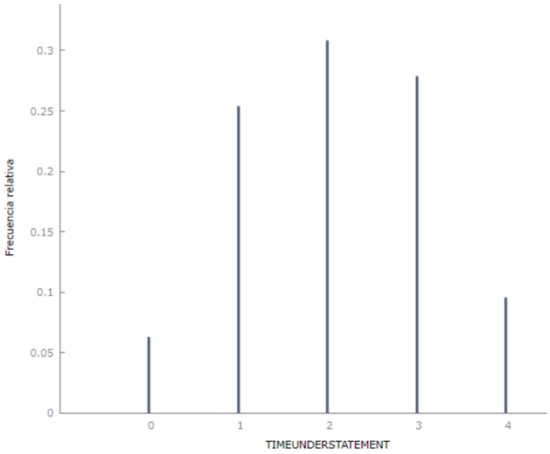
HOUSEHOLD MEMBERS DISTRIBUTION: DOGS DISTRIBUTION: CATS DISTRIBUTION:



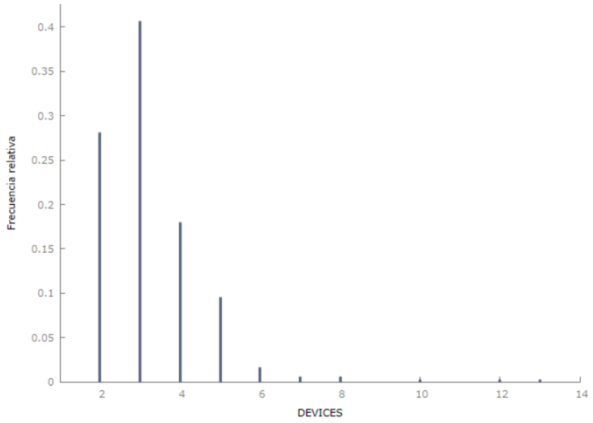
COUPLE DISTRIBUTION: IRRITABLE (ANNOYED) DISTRIBUTION: DEADLINES DISTRIBUTION:



UNDERESTIMATE (TIMEUNDERSTATEMENT) DISTRBUTION:

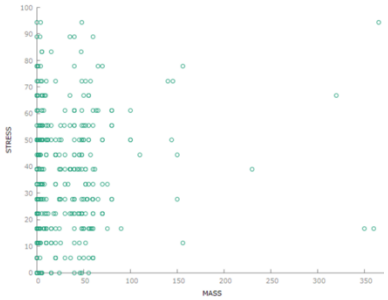


DEVICES DISTRIBUTION:

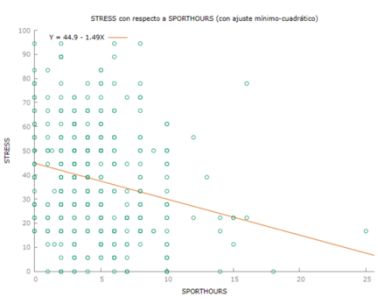


Scatter plots:

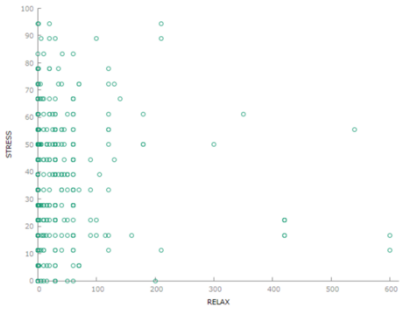
MASS vs. STRESS



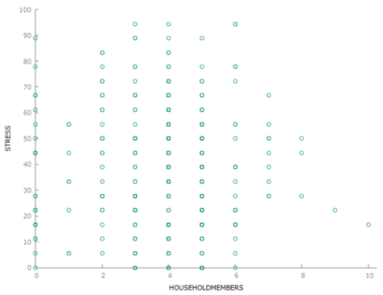
SPORTHOURS vs. STRESS



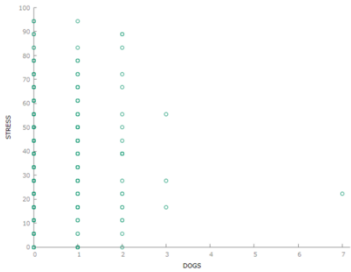
RELAX vs. STRESS



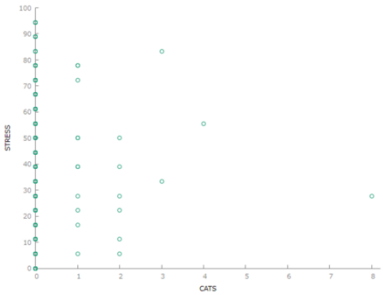
HOUSEHOLD MEMBERS vs. STRESS



DOGS vs. STRESS



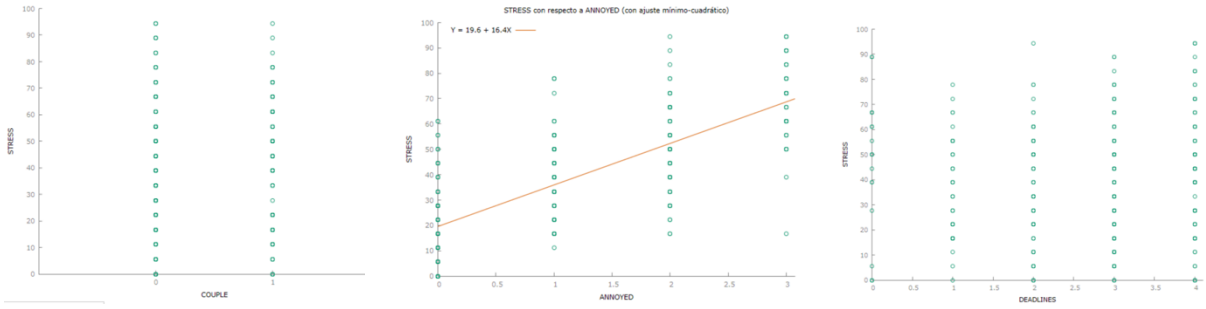
CATS vs. STRESS



COUPLE vs. STRESS

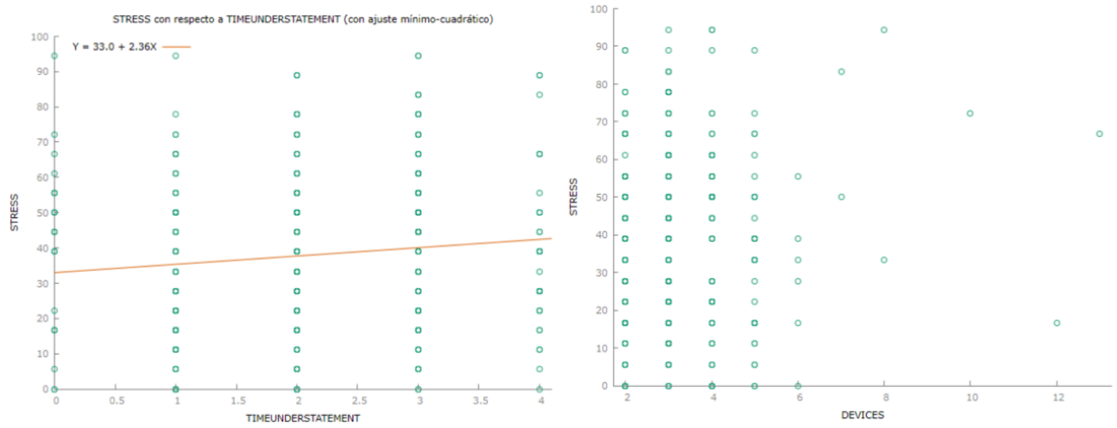
IRRITABLE vs. STRESS

DEADLINES vs. STRESS

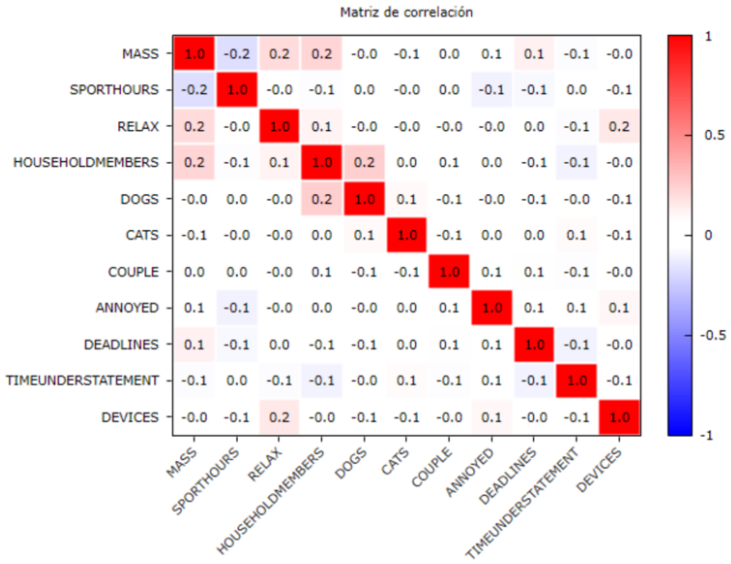


TIME UNDERTATEMENT vs. STRESS

DEVICES vs. STRESS



Correlation matrix between explanatory variables:



Initial exploratory analysis revealed that 8 out of 12 explanatory variables (MASS, SPORTHOURS, RELAX, HOUSEHOLDMEMBERS, DOGS, CATS, and DEVICES) contain potential outliers, identified using boxplots and standardized scores. While outliers are not automatically errors, their presence can significantly influence model estimates and reduce reliability. After seeing which model was the most optimal and efficient we decided to eliminate 5 explanatory variables that had outliers and 1 that did not until our model remained with only 6 of our initial 12 explanatory variables (SPORTHOURS, HOUSEHOLDMEMBERS, DOGS, COUPLE, IRRITATED, and TIMEUNDERSTATEMENT)

3.4.2 Justification of Transformations:

Transformations have not been incorporated into any variables from the model.

4. Research development and results

4.1 Estimated Models

Variable	M1	M2	M3	M4	M5
const	0,0164**	0,0056***	0,0043***	6,96e-11 **	1,04e-38***
Church	0,4209	0,3978	0,33984		
Sports	0,0665*	0,065*	0,0649*	0,0364**	0,0277**
Meditation	0,6065	0,6099	0,6104		
Housholds	0,92				
Dogs	0,9179	0,8936			
Cats	0,7109	0,7098	0,6997		
Couples	0,2132	0,2078	0,202		
Managment	0,3927	0,3951	0,3978		
perception	0,1857	0,1863	0,1871		
Devices	0,0524*	0,0521*	0,0523*	0,0859*	
R^2	0,034553	0,034525	0,034477	0,021247	0,013253
Adjusted R^2	0,007357	0,010117	0,01284	0,015855	0,010542

To analyse the significance of the various lifestyle related variables on students' self-perceived stress levels, we opted to use an approximated sequence of five OLS (Ordinary Least Squares) Regression Models, so the dependent variable in all models is the amount of irritableness or stress reported by students. The explanatory variables were used to evaluate their lone and all-together joint significance or therefore, lack of.

The objective of this progressive modelling strategy is to separate the effects of interest of the different variables and control for potential confounding factors.

After analysing the various models, we came to the conclusion that Model 4 is the best fit for the project as it confirms the importance of sports by the low P-value, which directly backs Hypothesis #2. It also points towards Devices being a potentially significant predictor. It helps to capture the significance of the individual-level perceptions without harming the model, not generating noise. Model 4 gave us a finer distinction of how student's lifestyles and stress perception which led us to choose it for a more in-depth interpretation.

4.2 Analysis of the chosen model

Collinearity check

```
Modelo 5: MCO, usando las observaciones 1-366
Variable dependiente: irritable

             coeficiente   Desv. típica   Estadístico t   valor p
  ------------------------------------------------------------------
  const       1.27473       0.0867468        14.69        1.04e-38 ***
  sports     -0.0332391     0.0150329        -2.211       0.0277   **

Media de la vble. dep.   1.120219   D.T. de la vble. dep.    0.988579
Suma de cuad. residuos   351.9829   D.T. de la regresión     0.983354
R-cuadrado               0.013253   R-cuadrado corregido     0.010542
F(1, 364)                4.888929   Valor p (de F)           0.027650
Log-verosimilitud       -512.1852   Criterio de Akaike       1028.370
Criterio de Schwarz      1036.176   Crit. de Hannan-Quinn    1031.472
```

To check the robustness of our Model, we used Model 4, firstly, we did a multicollinearity check. Multicollinearity happens when there's two or more than two independent variables in the regression model used, have a high correlation, which can lead to different results, one of them being an inflationary estimate of the coefficients, therefore making the model harder to analyse and interpret. This can be detected by using the Variance Inflation Factor (VIF). Usually, a VIF value greater than 5 (or, in more cautious analyses, greater than 10) indicates unwanted multicollinearity.

We conducted the VIF analysis for all explanatory variables included under Model 5, namely: Church attendance, Hours spent on sports, Meditation, Household size, owning dogs, Owning cats, Relationship status, Time management ability, Perception of stress, and Device usage. The VIF values we got were all under 5, indicating multicollinearity is not significant in the model. All the VIF values were well below 5, and this indicates that multicollinearity is not a big concern in our model.

This result supports that the variables are not highly collinear in their data. As an example, even intuitively one would expect variables such as Meditation and Church attendance to be collinear since they share our intuitions, our diagnostics suggest that they contribute uniquely to explaining stress percentage. In another hand, even though dogs and cats are both pet ownership, they have no collinear regression.

Similarly, even though both Dogs and Cats have a shared link with pet ownership, they are not collinear in this regression.

Overall, the absence of high multicollinearity makes our model more valid. It is the notification that each coefficient can be interpreted with greater confidence, since the variance of the estimates is not artificially inflated by inter-variable correlation. Therefore, the model assumptions are safe, and we can continue with the remaining diagnostic tests and interpretations on these results.

Heteroskedasticity test

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Contraste de heterocedasticidad de Breusch–Pagan
MCO, usando las observaciones 1–366
Variable dependiente: uhat^2 escalado

              coeficiente   Desv. típica  Estadístico t  valor p
  ------------------------------------------------------------------
  const        0.970004     0.304964         3.181       0.0016  ***
  church      −2.79854e−05  0.00129147      −0.02167     0.9827
  sports      −0.0119066    0.0166741       −0.7141      0.4756
  meditation  −0.000668158  0.000717151     −0.9317      0.3521
  housholds   −0.0366113    0.0357731       −1.023       0.3068
  dogs         0.0544360    0.0799465        0.6809      0.4964
  cats        −0.0320926    0.0949807       −0.3379      0.7356
  couple       0.108279     0.121287         0.8927      0.3726
  Managment   −0.0134585    0.0496991       −0.2708      0.7867
  perception   0.135343     0.0527510        2.566       0.0107  **
  Devices     −0.0135695    0.0425100       −0.3192      0.7498

  Suma de cuadrados explicada = 12.8548

Estadístico de contraste: LM = 6.427397,
con valor p = P(Chi–cuadrado(10) > 6.427397) = 0.778169
```

To analyse model 5's homoskedasticity assumption, we decided to use the Breusch-Pagan test, the standard test to check the heteroskedasticity in the linear regression models. Heteroskedasticity happens when the variance of error terms is heterogenous throughout all levels of explanatory variables, and it could make the efficiency of the estimates of the OLS (Ordinary Least Squares) as inefficient. The results show the test's output for the Breusch-Pagan test LM=6.427 is chi-squared with 10 degrees of freedom and a p-value=0.778, and as the p-value is significantly larger than the usual significance values, the null hypothesis of homoskedasticity cannot be rejected. By this we can conclude that, statistically, there's no signs of heteroskedasticity in Model 5. This output proves the constancy of variance assumption for the veracity of the regression estimates we got. No robust errors as our residual variance seems to be homoscedastic with all the observations.

Individual tests show that perception of stress contributes significantly to the variance of the residual that's p=0.0107. Which doesn't directly impact the collective contribution of the Breusch-Pagan test.

The model satisfies the homoskedasticity assumption, backing the inference from the OLS Estimation, which meant continuing with standard errors.

Variable significance

In Model 4, the variables "Sports" and "Devices" are proven to be as statistically important predictors at a level of 5%.

- *Sports* has a coefficient of 0.0277 which means $p < 0.05$. This supports the hypothesis that engaging in physical activity is positively and significantly linked with the dependent variable.
- *Devices* has a coefficient of 0.0532, also significant at the 5% level. This resulting that higher number of devices owned has a statistically significant positive effect, which might defy common expectations.

Whilst the next variables do not show statistical importance/significance: Religious Attendance, Meditation, Households, Pets (Dogs/Cats), and Couples. This is shown as their p-values exceed the 0.05 or 0.1 threshold.

We can conclude that only two variables in the model, shown clear statistical significance, and these will be the focus of our interpretation in the next part.

Interpret significant coefficients + fit measures (R^2, etc.)

The coefficient for Sports is 0.0277, which says that engaging in sports is associated with an increase of approximately 0.028 units in the dependent variable. This aligns with linking physical activity to improved well-being, self-perception, or quality of life, depending on how the outcome variable is defined.

The Devices variable has a coefficient of 0.0532, which also signifies that it's positively associated with the outcome. This may indicate that digital engagement has some beneficial component, maybe through social connection or productivity, but this interpretation would depend on the broader context of the study.

The R^2 for Model 5 is 0.0132, and the Adjusted R^2 is 0.0105, which means that the model explains just over 1% of the variance in the dependent variable. This is a relatively low value, this being common in cross-sectional studies on individual behaviour and psychology, where many other unobserved factors are at play. Still, even models with low R^2 values can give us meaningful insights if the coefficients are well-identified and significant.

The low R^2 suggests that, while sports and device usage are statistically significant, the overall explanatory power of the model is limited. Other models could be improved by adding more/other relevant variables.

Are the hypotheses supported or not?

1. **H1: Sports participation improves the outcome variable**: Supported. The variable Sports is statistically significant and positively associated with the dependent variable, aligning with the hypothesis.
2. **H2: Increased use of devices negatively affects the outcome**: Not supported. Contrary to what we expected, the Devices variable shows a positive and significant coefficient, which suggests that number of devices owned is associated with better outcomes in this context. This finding could reflect positive uses of technology, such as maintaining social ties or accessing mental health resources.
3. **H3: Attending church has a positive effect**: Not supported. The Church variable is not statistically significant, indicating no detectable relationship in this sample between religious participation and the dependent variable.
4. **H4: Meditation reduces stress/increases well-being**: Not supported. Like church attendance, Meditation is not a significant predictor, and thus this hypothesis is not confirmed in Model 5.
5. **H5: Pets (dogs/cats) are beneficial**: Not supported. Neither Dogs nor Cats shows a significant association with the outcome variable.
6. **H6: Family structure and management factors play a role**: Not supported. Households, Couples, and Management are all statistically insignificant.
7. **H7: Perception of stress is an important predictor**: Not supported. While Perception was significant in the heteroskedasticity test, it does not achieve significance in the main model.

In summary, out of all the proposed hypotheses, only the one related to sports participation is clearly supported by the data. Surprisingly, device use also appears beneficial, contrary to some expectations. The rest of the variables do not demonstrate statistically significant relationships, hence, not providing empirical foundation for the respective hypotheses.

4.3 Structural changes and exclusion tests

Structural Changes

To analyse potential structural change in the model, we conducted a Chow test using an augmented regression with interaction variables and group-specific slopes. The Chow test checks whether the relationship between the independent variables and the dependent variable (in this case, irritability) remains consistent across subgroups—typically before and after a potential structural break point. In this case, observation 183 was selected as the breakpoint.

```
Regresión aumentada para el contraste de Chow
MCO, usando las observaciones 1-366
Variable dependiente: irritable

                coeficiente   Desv. típica  Estadístico t  valor p
  ------------------------------------------------------------------
  const          1.03219      0.405336        2.547       0.0113  **
  church        -0.000274001  0.00173887     -0.1576      0.8749
  sports        -0.0292258    0.0226299      -1.291       0.1974
  meditation    -0.000913290  0.000925280    -0.9870      0.3243
  housholds     -0.0348611    0.0490370      -0.7109      0.4776
  dogs           0.0539224    0.119393        0.4516      0.6518
  cats          -0.0944338    0.163820       -0.5764      0.5647
  couple         0.0438436    0.174162        0.2517      0.8014
  Managment     -0.0167275    0.0646924      -0.2586      0.7961
  perception     0.0418304    0.0742102       0.5637      0.5733
  Devices        0.0871454    0.0624440       1.396       0.1637
  splitdum      -0.671445     0.576095       -1.166       0.2446
  sd_church      0.00218503   0.00244245      0.8946      0.3716
  sd_sports      0.000936830  0.0312801       0.02995     0.9761
  sd_meditation  0.00126346   0.00134268      0.9410      0.3474
  sd_housholds   0.0750735    0.0675617       1.111       0.2673
  sd_dogs       -0.0742017    0.153564       -0.4832      0.6293
  sd_cats        0.181212     0.194980        0.9294      0.3533
  sd_couple      0.132328     0.232909        0.5682      0.5703
  sd_Managment   0.118603     0.0934508       1.269       0.2052
  sd_perception  0.0463596    0.0996199       0.4654      0.6420
  sd_Devices    -0.0325159    0.0821953      -0.3956      0.6927

Media de la vble. dep.  1.120219   D.T. de la vble. dep.   0.988579
Suma de cuad. residuos  336.4652   D.T. de la regresión    0.988988
R-cuadrado              0.056755   R-cuadrado corregido   -0.000827
F(21, 344)              0.985646   Valor p (de F)          0.480544
Log-verosimilitud      -503.9341   Criterio de Akaike      1051.868
Criterio de Schwarz     1137.726   Crit. de Hannan-Quinn   1085.986

Contraste de Chow de cambio estructural en la observación 183
  F(11, 344) = 0.736109 con valor p 0.7036
```

The F-statistic for the Chow test is 0.736109, with an associated **p-value of 0.7036**. This result indicates that there is no statistically significant evidence of structural change at the chosen breakpoint. The high p-value (greater than the conventional 0.05 threshold) suggests that the regression coefficients are stable across the two subgroups (before and after observation 183), meaning the model behaves similarly over time or across the groups being compared.

Exclusion Tests

For the exclusion tests, the model has several interaction terms, named sd_variable, which are interactions between a split dummy variable and each of the original predictors. These assist in ascertaining whether the effect of each independent variable significantly varies between subgroups. All these interaction terms are not statistically significant (all p-values > 0.2), confirming the Chow test finding: there is no strong reason to assume that structural change exists in the model.

This reinforces the reliability of the model. Because no subgroup-related variations are detected, the effects of predictors on the outcome can be considered uniformly applicable to the entire sample since no variations are observed related to subgroups.

5. Conclusions

This study has analysed the effect of different lifestyle habits on stress levels among university students, using data collected from the “Database Models 2024–25”, comprising responses from ICADE students.

The research focused on three main hypotheses related to mindfulness, physical exercise, and church attendance.

After a progressive debugging of the model, a final model which offers a statistically and theoretically consistent interpretation of the relevant variables was selected. Among the variables selected only the

time spent doing sports and the number of digital devices owned where found to be statistically significant with stress levels. This supported one of our hypotheses that doing exercise helps reduce stress levels.

On the other hand, our other hypotheses were proven incorrect as the study did not find significant associations between stress levels and church attendance or meditation. Nevertheless, we did find out that the number of devices owned has a positive impact in the stress level of the students.

In conclusion while one of our hypotheses was supported (H2), the other related to mindfulness (H1) and religious attendance (H3) were not confirmed. The study's limitations include its cross-sectional nature, which does not allow for the establishment of casual relationships, and the potential of self-reporting bias.

For future research, it is recommended to consider model with longitudinal data for a better capturing the dynamics of stress and explore additional variables that may interfere with the student's well-being. Moreover, I believe that some strategies form the university promoting physical activity could help mitigate the stress of the students.